Event Set	PHASE OF FLIGHT	I/E ACTIVITY	EXPECTED OUTCOME
0	**SETUP**	**I/E ACTIVITY**	**EXPECTED OUTCOME**
	PVG-HKG **Spot 338:** **N31:07:51.65** **E121:47:16.83** **Hdg 256** **Routing:** **ZSPD HSN91D SHZ B221 LJG A470 DOTMI V512 ABBEY ABEY3B VHHH** **SIM 2** **FAA ID #1092** **SIM 3** **FAA ID #584** **SIM 4** **747-400 Version** **FAA ID #1091**	**POSITION SET:** AIRPORT: ZSPD RUNWAY: 35L Position: 338 (must use lat/long) **AIRCRAFT SET:** APU: Off Ground Power: EXT 1 connected and selected ZFW: 237.4 FUEL: 34.7 CG: 23.3 **ENVIRONMENT SET:** WIND: 260/19 TEMP: 21°C ALTIMETER: 1008 hPa RVR: NA SCT: 4500 tops 16000 VIZ: 10.0km TOPS: CLR TURBULENCE: N/A WX EFFECTS: N/A RUNWAY CONDITION: N/A VISUAL CONTROL: Actual time of day for sim event **For corrections or additions, please email: RLOFT747@AtlasAir.com**	Revision Changes: -Minor verbiage changes -Minor sequence changes

1	PREFLIGHT TO ENGINE START :30 (:30)	I/E ACTIVITY	EXPECTED OUTCOME
101	**After Prebrief, in briefing room:**	**Provide:** OFP/WX Paperwork w/o DDG Item, PF/PM role - Crew decides	Crew discussion 
102	**ATIS:** ZSPD DEP ATIS P XXXXz DEP RWY 35R AND 34L DCL/PDC in operation. Please acknowledge PDC. Caution for birds near the runway QNH 1008 Delivery FREQ 121.95 Wind 26010MPS, VIS 6000 HZ FEW Clouds 045, T 21 /DP 14, QNH 1008 HPA, Report receipt OF ATIS P on ZSPD	*[D-ATIS in printed pkg]* **Close to departure time, Maintenance informs crew:** *"The daily check has been accomplished. Aircraft Log is complete. The DDG that is on your release is the only one in the can."* **TRIGGER: Loadmaster informs crew**: *"We have added additional cargo, here is your loadsheet, am I cleared to close the door?"* *[Loadsheet in printed pkg]* **MINF ADJ should be accomplished.**	Crew discussion; MX, Dispatch coordination 
103	**IP Note:** Provide Loadsheet and FDP when asked	*[FDP in printed pkg]*	Crew discussion; MX, Dispatch coordination 

1	PREFLIGHT TO ENGINE START :30 (:30)	I/E ACTIVITY	EXPECTED OUTCOME
104	**IP Note**: Jeppesen page 20-1P6 states: **3.4 NOISE ABATEMENT** "Apply NADP1. If it cannot be implemented, inform ATC with a reasonable explanation."	**When crew requests clearance via PDC:** *[PDC in printed pkg]* **If crew requests ATC clearance via voice:** **Pudong Delivery:** *"Polar 482, Pudong delivery, cleared to Hong Kong via the HSN91D, then as filed. Climb initially 1800 meters, expect 10,400 meters one zero minutes after departure, departure on 120.3 Squawk 7225, runway 35R."* **Pudong Delivery:** *"Polar 482, read-back correct, contact Ground when ready for pushback and start clearance."*	The crew will perform a normal cockpit preparation, checklists and engine starts TME
105	**IP Note**: Use 'PB Left' key on IOS. Ensure 'P2' is at the aircrafts 1:30 position after pushback or use some PB straight to move it further back.	**When crew seeks clearance to push and start from Ground:** *"Polar 482, Pudong ground, stand 338 pushback and start approved, tail south, runway 35R for departure."* **When crew calls tug-driver:** *"Ground to cockpit, release brakes, commencing pushback, tail south"* ***[Pause]*** **"***Clear to start all engines."* **When pushback complete:** *"Ground to cockpit, pushback complete, set your brake."* ***[Pause]*** *"Tow bar disconnected, stand by for hand signals on the left side, have a good flight."*	Crew Briefing, Proper Cockpit to ground Interphone terminology TMP WM

2	ENGINE START THROUGH TAXI-OUT :10 (:40)	I/E ACTIVITY	EXPECTED OUTCOME
201		**TRIGGER: Before Taxi Complete**	Flaps set prior to A/C movement and ATC clearance to taxi 
202		**Taxi instructions as required from Ground:** **Pudong Ground:** *"Polar 482, taxi to runway 35R via P2, C, holding point P1."*	SOPs & Timing 
203	**IP Note:** Verify Runway Change Checklist completed FCOM guidance allow crews to complete the changes while moving; weather, traffic and Captains discretion Verify runway/trim verification completed	**TRIGGER: Aircraft is on Taxiway C** **Pudong Ground:** *"Polar 482, Runway 35R is now closed for maintenance, Runway is now 35L-HSN91D departure. Taxi to runway 35L via C, holding point P1."* **Pudong Ground:** *"Polar 482, you are number one, tower is 118.8, good-day"*	Required elements of Runway Change checklist complete. SOPs & Timing 

3	TAKEOFF THROUGH CLIMB :15 (:55)	I/E ACTIVITY	EXPECTED OUTCOME
301	**IP Note:** Observe runway verification procedure both in and outside. Altitude change for departure. 600m=2000ft	**Pudong Tower:** *"Polar 482, RNAV PD501, climb 600 meter, wind 260/10 mps, Runway 35L cleared takeoff"*	Crew flies RNAV TME AM
302	**IP Note:** Ref Jepp ZSPD (20-3E) Note 5: Crews should contact departure IMMEDIATELY without instruction once airborne	**TRIGGER: Only if crew requests frequency change** **Pudong Tower:** *"Polar 482, contact Pudong departure control 120.3."*	Departure Planning Briefed items
303	**IP Note:** 3600m=11,800ft Transition level=9,850ft Crew must read back "On Standard"	**Pudong Departure:** *"Polar 482, radar contact, climb via the HSN91D to 3600 meter on standard."*	Crew altitude awareness & discipline Read back of "on standard" to ATC TME V IA AM
304	**IP Note:** Sim 3 select "Generator Off-Over Excitation" then clear malfunction. EICAS will remain until QRH procedure is completed	**TRIGGER: Crew climbing on SID** Malfunction: ELEC GEN 2 OFF	Crew completes QRH procedure and manages Aircraft and Time accordingly PS

3	TAKEOFF THROUGH CLIMB :15 (:55)	I/E ACTIVITY	EXPECTED OUTCOME
305	**IP Note:** The FMC does not allow offsets on a published departure. The Crew must say unable. Continuous offset is available when direct to HSN or after. Crew must read back "On Standard" Change to VNAV Cruise Alt 7800m=25,600ft	**TRIGGER: At position NINAS** **Pudong Departure:** *"Polar 482, direct HSN, offset 3 miles Right of course, climb and maintain 7800m on standard, this will be your final altitude"*	Proper ATC terminology Accomplishes Route Offset Timing of tasks Crew communicates with dispatch a change in final altitude IA TME AM
306	**IP Note:** FOM 10.1.48 (more than 4000ft) Lower than planned cruise altitude forces a message or call to dispatch for a reanalysis. New expected fuel upon arrival is 12.5 if requested	**TRIGGER: 20nm prior to HSN** **Pudong Departure:** *"Polar 482, contact Hefei Control 118.97"* **Hefei Control:** *"Polar 482, radar contact"*	Proper ATC terminology GC PS

4	CRUISE TO DESCENT-DIVERT :15 (1:10)	I/E ACTIVITY	EXPECTED OUTCOME
401	**IP Note:** Change of VNAV Cruise Alt	**TRIGGER**: Approaching HSN **Hefei Control:** *"Polar 482, climb FL 260"*	Crew/Disp/ATC Coordination CD TMP WM
402	**IP Note:** [Note that this repo is not to a perfect spot but rather to a point from which we can resume the flight. This location is not important. Simply give the "cleared direct DOTMI" coming out of the flight freeze or reposition and move on with the scenario.]	**REPOSITION: After reaching FL260, wait three (3) minutes, then:** **Hefei Control:** *"Polar 482, fly heading 225"* Reposition simulator using Lat/Long: N23:11:00.00 E116:51:00.00 Hdg 225 Total Fuel 17.0	Proper ATC terminology
403		**Hefei Control:** *"Polar 482, direct DOTMI, Hong Kong Radar now 118.92 good-day"*	Proper ATC terminology
404		**Hong Kong Radar:** *"Polar 482, radar contact, expect ABBEY 3B arrival*." **Wait a few minutes:** **Hong Kong Radar:** *"Polar 482, cross MUSEL at FL130"*	Proper ATC terminology TMP AM

5	DESCENT-DIVERT TO ARRIVAL :20 (1:30)	I/E ACTIVITY	EXPECTED OUTCOME
501	**Configure IOS for HKG arrival weather, runway 25L:** **VHHH Arrival ATIS:** HONG KONG Arrival Information T at time XXXXz Arrival Runway 25R, departing 25C, Wind 190 DEG 18 KT, visibility 10 KM smoke mist Cloud OVC 5000FT, Temperature 18 Dewpoint 17, QNH 1014 HPA Acknowledge Information T on first contact with Approach. Crew should advise Dispatch of holding intent and time.	**TRIGGER: At position FISHA** **Hong Kong Radar:** *"Polar 482, due to traffic, I have holding instructions when ready to copy"* ***The EFC time should be <u>10 minutes</u> from waypoint crossing time*** **Hong Kong Radar:** *"Polar 482, hold at ABBEY, as published except left hand turns, EFC time XXXXz, descend and maintain FL130"* *[D-ATIS in printed pkg]* *[FDP in printed pkg]*	Proper ATC terminology Crew/Dispatch Coordination PS TMR
502	**IP Note:** **In icing conditions**. Crew should use best speed hold in clean configuration. If "INSUFFICIENT FUEL" msg is shown, one option is to adjust EFC momentarily to current time to display "HOLD AVAIL" information. Crew could decide to amend/delete flight plan reserve fuel figure and incorporate current weather into their decision to not proceed to an alternate. **Hold speed FL130: 230kt MAX**	**TRIGGER: Crew reports entering the hold** **Hong Kong Radar:** *"Polar 482, what is your hold time available?"* ***Crew references the "Hold Available" selection of the FMC Hold page*** **Hong Kong Radar:** *"Polar 482, roger"*	Crew Briefing, Dispatch Coordination TMP TMR

6	ARRIVAL TO APPROACH :10 (1:40)	I/E ACTIVITY	EXPECTED OUTCOME
601	**IP Note:** FMC may generate a left turn to TEDUR, crew should use HDG SEL to the right until FMC generates the correct turn as per ATC instructions	**TRIGGER: After 3 turns in the hold or crew declares "Minimum Fuel"** **Hong Kong Radar:** *"Polar 482, **right turn** direct TEDUR, ABBEY 3B arrival"* **[Wait until direct TEDUR]** **Hong Kong Radar:** *"Polar 482, contact Hong Kong Approach 121.3"*	Crew coordination Proper ATC terminology AM TME CD LF
602	**Runway 25R** Transition Level FL110 VNAV DES forecast page displays FL090 as standard, crew must manually change their transition level	**TRIGGER: Crew calls Hong Kong Approach** **Hong Kong Approach:** *"Polar 482, ILS runway 25R RIVMI Transition"* **TRIGGER: 20nm prior to TEDUR or when crew requests descent** **Hong Kong Approach:** *"Polar 482, descend via the RIVMI transition, QNH 1014 hPa."*	Proper ATC terminology Altitude awareness TME EM

7	APPROACH through Landing :20 (2:00)	I/E ACTIVITY	EXPECTED OUTCOME
701	**IP Note:** Headwind Component: 8 kts Vref Additive: 5 kts	**TRIGGER: Passing RUNSU** **Hong Kong Approach:** *"Polar 482, From TOPUN, Cleared ILS 25R, Contact Tower at TOPUN 118.7"*	Proper ATC terminology
702	**IP Note:** This is designed as an early go-around once established on the approach. Crews can utilize either the TOGA switch or AP/AT off, both FDs off then on to decouple the LOC/GS if captured. Emphasis is on positive control of the aircraft and altitude awareness.	**TRIGGER: At waypoint SAGNI and descending on the arrival:** **Hong Kong Approach:** *"Polar 482, traffic, go around, fly the published missed approach. Contact Hong Kong Director 119.5"* **Hong Kong Director:** *"Polar 482, radar contact, expect ILS 25R BOKAG transition"*	Aircraft coordination Altitude crossing restriction adherence 
703	**IP Note:** Complex missed approach procedure	**TRIGGER: Approaching VH526** **Hong Kong Director:** *"Polar 482, speed at your discretion, cleared ILS 25R BOKAG transition."*	
704	**IP Note:** Headwind Component: 8 kts Vref Additive: 5 kts	**TRIGGER: Passing RUNSU** **Hong Kong Director:** *"Polar 482, contact tower 118.7"* **Hong Kong Tower:** *"Polar 482, wind 190 at 18, Runway 25R cleared to land"*	Crew Normal Procedures 

ALT	**IP Note:** If another airport or course of action is taken, react and coordinate accordingly. Not all scenarios can be scripted. **Diversion Only**	**TRIGGER: Diversion or alternate course of action requested** ***Leave weather as planned prior due to close nature of other airports*** **Shenzen ZGSZ** is 20 NM due north. ATC would vector well east prior to turning into China. Recommend NEDLE then vectors to final. Weather is sufficient for RWY 15 and 16 **Macao VMMC** is 20 NM Southwest. ATC would vector well out over the water prior to turning toward Macao. Weather is sufficient for RWY 16 (LOC or RNAV only). Rwy 34 exceeds maximum tailwind by 1kt. **Reposition the VHHH holding short 07R after landing is complete**	Crew Normal Procedures 
8	**TAXI-IN THROUGH POSTFLIGHT :10 (2:10)**	**I/E ACTIVITY**	**EXPECTED OUTCOME**
801		**TRIGGER: Exiting Runway** **Hong Kong Tower:** *"Polar 482, Right turn left on Charlie, hold short of Yankee, Ground 121.6."*	Crew Vigilance 
802	**IP Note:** This is the end of Leg 1. Configure IOS for next leg and leave engines running. **Reset BRAKE TEMP and clear any malfunctions**	**Advise Crew:** Taxi down taxiway Charlie, then come to a stop. This is the end of Leg 1. We will resume from this position after the break. **Reposition the VHHH holding short 07R after scenario is complete**	

Event Set/Script Index	PHASE OF FLIGHT	I/E ACTIVITY	EXPECTED OUTCOME
0	**SETUP**	**I/E ACTIVITY**	**EXPECTED OUTCOME**
	HKG-ANC **Routing:** **VHHH DALOL _A DALOL V631 ENVAR PUTER A590 HAMND NEELL6 PANC** **SIM 2** **FAA ID #1092** **SIM 3** **FAA ID #584** **SIM 4** **747-400** **FAA ID #1091**	**POSITION SET:** AIRPORT - VHHH RUNWAY 07R Holding Short **AIRCRAFT SET:** APU: Off ZFW: 270.5 FUEL: 126.3 CG: 20.6 **ENVIRONMENT SET:** WIND: 120/10 Gust 25 **(FAA Gust)** TEMP: 18°C ALTIMETER: 1014 hPa RVR: N/A SCT: 5000' VIZ: 10.0km TOPS: CLR TURBULENCE: Light WX EFFECTS: N/A RUNWAY CONDITION: N/A VISUAL CONTROL: Actual time of day **For corrections or additions, please email: RLOFT747@AtlasAir.com**	Revision Changes: -DALOT Departure change and extension -Updated to new OFP -Climb restriction focus until SEGPO

1	PREFLIGHT TO ENGINE START	I/E ACTIVITY	EXPECTED OUTCOME
101	**IP Note:** Select FAA Gust **VHHH departure ATIS:** HONG KONG Arrival Information T at time XXXXz Arrival Runway 07C, departing 07R, Wind 120 DEG 10 KT, Gusting to 25, visibility 10KM smoke mist Cloud OVC 5000FT, Temperature 18 Dewpoint 14, QNH 1014 HPA Acknowledge Information T on first contact.	Provide OFP/WX Paperwork *[D-ATIS in printed pkg]* *[PDC in printed pkg]* **Close to departure time, Maintenance informs crew:** *"The Transit check has been accomplished. Aircraft Log is complete. The DDG that is on your release is the only one in the can."* **TRIGGER: Loadmaster informs crew**: *"Here is your loadsheet and NOTOC, am I cleared to close the door?"* *[Loadsheet and NOTOC in printed pkg]* *[FDP in printed pkg]*	Crew discussion, MX, Dispatch coordination 
102	**IP Note:** Position the aircraft to Hold short of runway 07R, flaps up, engines running, RTO selected. Inform the crew they have all engines running, wave-off/salute has been received, flaps & flight control check must still be accomplished and to pick up their normal responsibilities there.	**TRIGGER: If not in position after previous leg…** Reposition on taxiway short of runway 7R. Strong gusty X-Winds present from mountain side. Or **Hong Kong Tower:** *"Polar 483, taxi to runway 07R holding point K1, you are number 1 for departure"*	

2	ENGINE START THROUGH TAXI-OUT :30 (:30)	I/E ACTIVITY	EXPECTED OUTCOME
201	**IP Note:**	**TRIGGER: Takeoff Before Takeoff Checklist complete.**	Runway and Trim Verification SOPs 
202	**IP Note:** Adjust fuel to above min fuel (124.6) to account for engines running during preparations. **NP.35.1**: Use a minimum amount of runway when aligning the airplane for takeoff when the stop margin is less than 1000 ft.	**Hong Kong Tower:** *"Polar 483, Hong Kong tower line up and wait runway 07R."*	Crew Normal Ops, SOP & Timing Proper ATC Terminology 

3	TAKEOFF THROUGH CLIMB :05 (:35)	I/E ACTIVITY	EXPECTED OUTCOME
301	**IP Note:** Confirm runway/trim verification	**TRIGGER: Line up and wait** **Hong Kong Tower**: *"Polar 483, wind 120/10 Gust25 runway 07R, RNAV PORPA, Cleared for takeoff."*	Proper ATC Terminology Crew flies RNAV TME
302		**TRIGGER: Airborne** **Hong Kong Tower:** *"Polar 483, Hong Kong departure 123.8 good day."*	Proper ATC Terminology TME TMR V CD
303		**Hong Kong Departure:** *"Polar 483, radar contact, maintain one four-thousand, high-speed approved."*	PS WM

4	CRUISE TO DESCENT-DIVERT :15 (0:50)	I/E ACTIVITY	EXPECTED OUTCOME
401	**IP Note:** Expect the crew to declare a MAYDAY due to the fire and make some kind of request.	**TRIGGER: Passing SEGPO** **IP Selection:** **FIRE MAIN DECK (Sim 2)** **FIRE AFT CARGO (Sim 3)**	
402	**IP Note:** FOB should be in minutes vs KG This script expects, and is written for, a decision to air turn back to HKG. Another course of action may not be wrong, however due to simulator limitations it is the most practical/realistic.	**TRIGGER: Crew declares MAYDAY** **Hong Kong Departure:** *"Mayday Polar 483, Direct position LIMES, descend 6000, expect ILS 07R"* After response **Hong Kong Departure:** *"Mayday Polar 483, Say your estimate for LIMES, and confirm souls and fuel on board, confirm squawk 7700"*	Crew/ATC coordination, Timing CM
403	**IP note:** It is not expected for the Crew to contact Dispatch. If Crew does initiate contact, be brief and offer assistance.	If Crew calls Dispatch, obtain their current fuel on board.	Crew/ Dispatch coordination CD TMP
405	**ATIS:** HONG KONG Arrival Information Whiskey at time XXXXz Arrival Runway 07C/07R, Wind 120 DEG 15KT, visibility 4KM smoke mist OVC 1100FT, Temperature 18 Dewpoint 14, QNH 1014 hPa Acknowledge Information Whiskey on first contact with Approach.	*[D-ATIS in printed pkg]* *[FDP in printed pkg]* **Overweight Landing is required**	TMP TMR

5	DESCENT-DIVERT TO ARRIVAL :10 (1:00)	I/E ACTIVITY	EXPECTED OUTCOME
501	**IP Note:** QRH FIRE MAIN DECK: Step 14 Note: Maintain VMO/MMO until 15nm from the runway	**TRIGGER: Heading direct LIMES** **Hong Kong Departure:** *"Mayday Polar 483, contact director 119.5"* **Hong Kong Director:** *"Mayday Polar 483, cross LIMES at or above 3000, cleared ILS 07R via LIMES"*	CM TME TMR GC
6	**ARRIVAL TO APPROACH** **:10 (1:10)**	**I/E ACTIVITY**	**EXPECTED OUTCOME**
601		**TRIGGER: Passing LIMES** **Hong Kong Director:** *"Mayday Polar 483, call Hong Kong tower now 118.4"*	TMP TME V

7	APPROACH through Landing :10 (1:20)	I/E ACTIVITY	EXPECTED OUTCOME
701	**IP Note:** **Make note of any stabilized approach criteria exceedances that are not called out/corrected** **CFR will query a/c on estimated stopping position to better position themselves for emergency access**	**TRIGGER: ATC Handoff** **Hong Kong Tower:** *"Mayday Polar 483, Hong Kong Tower, Wind 120 at 15, runway 07R, cleared to land."* **Hong Kong Tower:** *"Mayday Polar 483, say estimate stopping position"*	PM assertiveness should the approach de-stabilize. 
702	**IP Note:** **QRH- FIRE MAIN DECK:** Step 17: AUTOBRAKES selector… MAX AUTO	**TRIGGER: On touchdown** **Hong Kong Tower:** *"Attention all Hong Kong area traffic, runway 07R is closed, repeat runway 07R is closed. BREAK Mayday Polar 483 heavy, hold position on the runway, Fire Rescue Equipment in transit. Monitor 122.95. The fire chief will communicate with you on this frequency.* **Fire Chief:** *"Polar 483, this is Fire Chief One...how do you read?"* **Fire Chief:** *"Polar 483, confirm your approximate position on Runway 07R?"* **Fire Chief:** *"Polar 483, Standby for further instructions, we are approaching your aircraft now"*	Crew monitors descent rate at touchdown for MX report 

8	TAXI-IN THROUGH POSTFLIGHT :05 (1:25)	I/E ACTIVITY	EXPECTED OUTCOME
801	**IP Note:** Give the crew time to accomplish the situation or emergency. Once complete, advise them we are complete with the RLOFT and are going to accomplish **2 additional takeoff and landings each along with the SPOT training**. Configure IOS and setup using a sim reset. If any CBs are pulled to comply with DDG, reset them prior to landing currency. If there is seat support, only student landing currency is necessary. **Requirements:** • **Takeoff Departure Stall** • **Clean Configuration Stall** • **Landing Configuration Stall** • **KJFK-VOR 31L Approach** • **Visual Approach Pattern** • **Crosswind Landing (25+ kts)** • **Landing Currency**	**IP Note:** **The RLOFT session is complete, the remainder of this session is SPOT training. Reposition to KJFK.** Complete the **JFK-VOR 31L approach**. Emphasis is on the visual approach segment along with stabilized approach criteria. This approach has one of the highest rate of unstabilized approaches at our airline and others. Recommend winds 220/10-15 knots. For landing currency, **a visual pattern and manual flight is required**. This is to promote manual flying skills and pitch/power control as emphasized in FAA SAFO 13002. The instructor is encouraged to review pitch, power, energy management, and stabilized approach criteria. Discuss **Pilot Induced Oscillation (PIO), hard landings and strategies to mitigate these issues**. It is recommended that the visual approach landing is made with **25+ knots of crosswind** to emphasize the PIO and hard landing mitigation strategies. Baulked landings/low energy go-around versus full stop landings are acceptable and at instructor discretion such that students leave with 3 takeoffs and 3 landings each (total). Main landing gear must touch the runway, straddling the centerline, to count as a landing and subsequent takeoff. During these approaches, complete the SPOT training requirements of a **takeoff, clean configuration, and landing configuration stalls**. Recovery can be initiated **at buffet (natural) or stick shaker (artificial)**. Once complete, instruct them to taxi to park, and perform a shutdown and secure checklist. The words “Secure Checklist Complete” signify the conclusion of the simulator period.	